

SIMON KAMUA

PLC/SIS TECHNICIAN | ICSS, INSTRUMENTATION & PROCESS AUTOMATION

Buliisa, Uganda | +256 775 497 383 | kamuasimeon@gmail.com | [LinkedIn](#) | <https://simonkamua.github.io/index.html>

PROFESSIONAL PROFILE

PLC/SIS Technician with a First-Class BSc in Chemical Engineering and progressive TotalEnergies Tilenga Project experience spanning systems administration, field instrumentation, commissioning, and integrated control and safety systems. Combines process-engineering insight with complete-loop troubleshooting, ICSS testing, and analysis of equipment behaviour. Traces pressure, temperature, flow and level measurements from field sensors through I/O and controller logic to alarms, permissives, interlocks, SIF actions and final elements. Brings existing facility knowledge and structured, safety-focused fault diagnosis to commissioning, start-up and long-term plant operations.

VALUE PROPOSITION

- **Analytical troubleshooting:** Traces abnormal behaviour across field devices, wiring, I/O, communications, controller logic and final elements instead of treating symptoms in isolation.
- **Process-automation integration:** Uses Chemical Engineering knowledge to relate control-system behaviour to process safety, stability, equipment protection and production availability.
- **Safety and reliability mindset:** Documents findings, identifies technical risk, and supports evidence-based testing and restoration under PTW, isolation and HSE controls.
- **Project continuity:** Existing knowledge of Tilenga facilities, work processes, interfaces and operating constraints supports continuity from commissioning into long-term operations.

CORE TECHNICAL COMPETENCIES

Control and safety systems: PLC, SIS and ICSS operation, monitoring, testing and first-line troubleshooting; alarm, permissive, interlock, sequence and shutdown-function verification

Testing and commissioning: Point-to-point checks, loop validation, signal simulation, functional testing, cause-and-effect verification, punch-list support and handover readiness

Functional safety: SIF sensor-logic-final-element tracing; voting, trip setpoint, fail-safe action, bypass/inhibition, reset and restoration awareness

Reliability analysis: Alarm/event correlation, complete-loop diagnosis, structured fault isolation, defect reporting, root-cause thinking and restoration verification

Engineering documentation: P&IDs, loop and wiring drawings, logic diagrams, control narratives, cause-and-effect matrices, datasheets and OEM manuals

Field instrumentation: Pressure, temperature, flow and level measurement; Fire and Gas; condition monitoring; solenoid-operated valves and control valves

PROFESSIONAL EXPERIENCE

PLC/SIS Technician

May 2026-Present

TotalEnergies EP Uganda - Tilenga Project

Buliisa, Uganda

- Analyze pressure, temperature, flow and level conditions alongside process trends, alarms, field feedback and equipment status to distinguish process deviations from instrumentation, I/O, communication and control-logic faults.
- Trace control and protective functions from process measurements through ICSS logic to alarms, permissives, interlocks and final-element actions, confirming equipment response against the intended operating philosophy.
- Develop Maintenance Operation Procedures (MOPs) for ICSS, PLC and SIS activities, translating system documentation, OEM guidance and process requirements into controlled maintenance sequences covering prerequisites, operational risks, isolations and inhibitions, execution steps, functional testing, recovery actions and safe return to service.
- Apply P&IDs, control narratives and cause-and-effect matrices to evaluate process sequences, equipment dependencies, shutdown actions and safe restoration requirements.
- Support commissioning and start-up readiness by evaluating interactions among pumps, compressors, valves, package systems and their upstream and downstream process conditions.
- Assess the process consequences of alarms, overrides, inhibitions and equipment unavailability before testing, restoration or return to service.
- Document system behaviour, test results and technical findings to support fault resolution, commissioning knowledge retention and continuity into steady-state plant operations.

Selected Process and System Exposure

- **Wellpad production systems:** Well control, production routing, skid interfaces and monitoring of conditions affecting safe hydrocarbon production.
- **Annulus gas compression and rotating equipment:** Compressor start-up sequences, permissives, interlocks, trips, VSD interfaces and equipment-status feedback.
- **Multiphase flow measurement and pilot test separation:** Measurement and validation of oil, gas and water production rates for well-performance monitoring.
- **Polymer injection and water-intake facilities:** Control and monitoring of water supply, treatment and polymer dosing supporting injection operations.
- **Fire and Gas and HIPS:** Detection, voting, shutdown and pressure-protection functions safeguarding personnel, equipment and process facilities.

Instrumentation and Control Technician

Apr 2024-Apr 2026

TotalEnergies EP Uganda - Tilenga Project

Buliisa, Uganda

- Installed, inspected, calibrated, pre-commissioned, commissioned and maintained field instrumentation across central processing facilities, wellpads, crude-oil pipeline systems and water-intake facilities.
- Calibrated and validated Rosemount and Yokogawa pressure, temperature, flow and level instrumentation, relating measurement ranges, accuracy and response to associated process conditions and equipment-operating requirements.
- Applied Chemical Engineering knowledge of fluid flow, pressure-temperature relationships, phase behaviour and process-system interactions to interpret abnormal readings, evaluate instrument suitability and distinguish genuine process deviations from instrumentation faults.
- Inspected and tested Fisher control valves, positioners, actuators and solenoid-operated valves, verifying travel, feedback, response, air supply and fail-safe action while assessing effects on process stability, isolation and equipment protection.
- Supported testing of Det-Tronics Fire and Gas devices, condition-monitoring instrumentation and Safety Instrumented Functions, tracing protective actions from field sensors through control-system interfaces to alarms, trips and final-element response.
- Performed installation verification, continuity and point-to-point testing, signal simulation, loop checking and functional validation, confirming the interdependence of field measurement, signal transmission, control logic and equipment response.
- Supported functional-safety integrity by verifying SIF sensor and final-element operation, trip setpoints, voting inputs, fail-safe behaviour and correct restoration after testing, maintenance or inhibition in accordance with approved procedures and assigned SIL requirements.
- Conducted preventive and corrective maintenance and systematic fault-finding across complete instrument loops, diagnosing calibration drift, impulse-line problems, wiring and power-supply faults, signal instability, communication failures, air-supply problems and control-valve response deviations.
- Interpreted P&IDs, loop and wiring diagrams, hook-up drawings, datasheets, cause-and-effect matrices and OEM manuals to understand each instrument's process purpose, control function, protective role and consequence of failure.
- Coordinated with operations, control-room, commissioning and engineering teams during testing, fault resolution, restoration and system handover under approved PTW, isolation, hazardous-area and HSE requirements.

Selected Instrumentation Exposure

Rosemount and Yokogawa pressure, temperature, flow and level instrumentation; Det-Tronics Fire and Gas devices; condition-monitoring instruments; Fisher control valves, positioners and actuators; and solenoid-operated valves.

Systems Administrator

Jan 2023-Feb 2024

Kyambogo University

Kampala, Uganda

- Installed, configured and maintained Windows and Linux systems, servers and network infrastructure, supporting reliable and secure organizational operations.
- Administered Active Directory, DNS, DHCP, VPN services, user accounts and access permissions, developing a practical understanding of networked-system architecture, device communication and controlled access.
- Monitored system and network performance and applied structured fault diagnosis across hardware, operating systems, applications and communication paths to identify root causes and minimize service interruption.
- Implemented security controls including firewall configuration, antivirus management, system hardening, patching and access control, establishing a foundation for industrial control-system cybersecurity and secure system administration.
- Performed backup, recovery and data-integrity checks while supporting virtualization and storage systems; maintained configuration records, troubleshooting procedures and incident documentation to strengthen availability, controlled change and repeatable fault resolution.

- Collaborated with users and technical teams during incident resolution and infrastructure upgrades, translating operational requirements into technical solutions and developing transferable competence in networking, communication protocols, cybersecurity and availability management for instrumentation, ICSS and PLC/SIS responsibilities.

SELECTED COMMISSIONING AND TECHNICAL CONTRIBUTIONS

- **NGR-01 annulus gas compressor:** Supported commissioning by evaluating interfaces among process instrumentation, WHCP/HPU, UCP and ICSS indications, with attention to equipment feedback, permissives, interlocks, alarms and start-up readiness.
- **HIPS-102, HIPS-103 and inhibition management:** Reviewed preventive HIPS functions and inhibition requirements by tracing protective actions across sensors, logic conditions and final elements while considering the process consequences of inhibiting, testing and restoring safety functions.
- **JBR-04 wellpad systems:** Developed system understanding across the ISS skid, MPFM, sand-jetting system, pilot test separator and polymer-injection facilities, relating production objectives to measurements, control sequences, equipment dependencies and protective actions.
- **Rotating equipment and supporting systems:** Evaluated the interaction of ABB VSD-controlled equipment, UPS, earthing and cathodic-protection systems with control-system availability, equipment protection and reliable plant operation.

SELECTED AUTOMATION PROJECTS

- **Water-Tank PCS and SIS Control System - In development:** Designing Omron PLC level control using a modulating inlet valve, VFD-controlled duty/standby pumps and automated bumpless changeover. Developing an independent ESP32 educational SIS demonstrator using three level transmitters with 2oo3 voting, trip logic and fail-safe final actions. Deliverables include a control philosophy, P&ID, I/O list, cause-and-effect matrix, fault handling, simulations and test procedures.
- **Automated Paint-Production Plant - Completed simulation:** Developed state-based batch-control logic covering material charging, mixing, timed stages, permissives, interlocks, pause/abort behaviour and coordinated equipment operation.
- **Compressor Start-Control Sequence - Completed simulation:** Programmed equipment permissives, sequential start-up, protective interlocks, trip handling, first-out indication and reset conditions, with attention to safe equipment response under abnormal conditions.
- **Multi-Zone Conveyor Control System - Completed simulation:** Developed sensor-based material tracking, four-zone coordination, queue management, jam detection and interlocking to prevent conflicting movements and support orderly transfer.

Portfolio note: Projects are independently developed educational/simulation work. The ESP32 SIS is not a certified logic solver and is not suitable for operational SIL service.

EDUCATION

Bachelor of Science in Chemical Engineering - First-Class Honours

Kyambogo University

CGPA 4.45/5.00

Kampala, Uganda

Relevant foundation in process control, fluid mechanics, thermodynamics, mass and energy balances and process safety. This supports current interpretation of process behaviour, instrumentation performance, control logic, final-element response and equipment interactions, while providing a strong base for future PLC/ICSS engineering and supervisory responsibilities within plant operations.

CERTIFICATIONS AND TECHNICAL TRAINING

- OPITO Global Level 2 - Maintenance Practice (Instrument and Control), SCQF Level 5; Institut Teknologi Petroleum PETRONAS (INSTEP).
- TotalEnergies Customized Training Programme for the Tilenga Project - Instrumentation Trade; INSTEP.
- OPITO Global Level 3 - Instrumentation and Control - In progress.
- OPITO Global Foundation: Introduction to Oil and Gas Processing; Introduction to Maintenance Workshop Practice, SCQF Level 3.
- ATEX Level 1; Electrical Clearance B1V-BV-BR-BE-BC-H0-H0V; On-site Work Management; Permit to Work; Confined Spaces; Practice of Golden Rules.
- BOSIET with EBS; CA-EBS Initial Deployment; Self-Contained Breathing Apparatus; Firefighting; Basic First Aid.
- Google IT Support Professional Certificate - operating systems, networking, cybersecurity and structured digital-system troubleshooting.

ACHIEVEMENTS AND RECOGNITION

- **PETRONAS Focused Recognition - Customer Focused and Courage to Act | INSTEP, Jun 2025:** Recognised for voluntarily troubleshooting and restoring three Allen-Bradley ControlLogix simulators, ensuring their availability for continuous PLC training delivery.
- **Best Instrumentation and Control | TotalEnergies:** Awarded for outstanding technical understanding and practical performance during specialist Instrumentation and Control training.
- **Career progression:** Progressed from Instrumentation and Control Technician to PLC/SIS Technician in May 2026, extending responsibility from field devices and complete loops to integrated control and safety systems.

PROFESSIONAL MEMBERSHIPS AND DEVELOPMENT PATHWAY

Professional Memberships

- **International Society of Automation (ISA):** Professional Member, July 2026-Present; engagement with recognised practices in industrial automation, process control, functional safety and control-system cybersecurity.
- **Uganda Institution of Professional Engineers (UIPE):** Graduate Member, GM/5319, April 2026-Present; continued professional development toward recognised engineering practice in Uganda.

Planned Professional Development

- **ISA Certified Automation Professional (CAP):** Planned validation of competence across control-system design, integration, commissioning, maintenance and lifecycle management.
- **IEC 61511 Functional Safety certification:** Planned deeper capability in the SIF lifecycle, SIL verification, proof testing, bypass control and SIS performance during plant operations.
- **IEC 62443 Industrial Cybersecurity certification:** Planned competence in secure administration, segmentation, access control and protection of ICSS and operational-technology systems.
- **CompTIA Network+:** Planned reinforcement of network troubleshooting, device communication and control-system infrastructure support.

SELECTED SYSTEMS AND PLATFORMS

ICSS and safety systems: Foxboro DCS; HIMA Planar4 and HIMatrix

PLC platforms: Allen-Bradley ControlLogix; Siemens S70 and Logo; Omron PLCs; CODESYS simulation environment

RTU and communications: ABB RTU560; industrial Ethernet; networked-device communication and control-system interface diagnostics

Machinery protection: Baker Hughes Bently Nevada 3500 condition-monitoring systems

Process measurement and monitoring: Emerson Roxar MPFM and SAM; Endress+Hauser; FEBUS fibre-optic monitoring systems

Electrical and availability interfaces: Gutor UPS systems; Basler DECS-250 excitation control; power-supply monitoring and control-system availability interfaces

Embedded and educational development: ESP32/Arduino programming; 2oo3 voting demonstrations; sequential control and fault-handling simulations

Systems administration: Windows and Linux; Active Directory; TCP/IP; DNS; DHCP; VPN; access control; backup and recovery

PROFESSIONAL LINKS AND REFERENCES

LinkedIn: [linkedin.com/in/simon-kamua-145562236](https://www.linkedin.com/in/simon-kamua-145562236)

Automation portfolio: Engineering dossiers, control philosophies, diagrams, PLC logic, interactive simulations and test evidence available upon request; public portfolio site in development

Professional and technical references are available upon request.